

Project Report: SMS Spam Detection Using Machine Learning

Project Title: SMS Spam Detection Using Machine Learning

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2. Introduction

In the modern digital era, Short Message Service (SMS) remains a popular communication tool. However, the convenience of SMS is often marred by the prevalence of spam messages, which can range from harmless advertisements to malicious phishing attempts. SMS spam not only inconveniences users but also poses security risks. Traditional keyword-based filters are often

inadequate due to the evolving nature of spam. This project explores the use of machine learning to create an adaptive and intelligent spam detection system.

3. Problem Statement

SMS spam is a growing concern, characterized by its ability to bypass simple filters and reach users' inboxes. The challenge lies in accurately distinguishing between legitimate personal messages and spam, especially when spam messages mimic the style of ham. The project addresses the need for a highly accurate and efficient classification system that can handle the linguistic nuances of SMS text.

4. Objectives

The primary objectives of this project are:

- To clean and preprocess the SMS Spam Collection dataset for machine learning.
- To conduct Exploratory Data Analysis (EDA) to identify patterns in spam vs. ham messages.
- To implement advanced NLP techniques, including tokenization, stop-word removal, and stemming.
- To compare the performance of multiple classification models (Naive Bayes, Logistic Regression, SVM).
- To evaluate the models using accuracy, precision, recall, and f1-score.

5. Scope of the Project

The scope of this project includes the development of a complete machine learning pipeline for SMS spam classification. It covers data acquisition, preprocessing, feature engineering using TF-IDF, model training, and performance evaluation. The project focuses on supervised learning techniques and does not include the deployment of the model as a mobile application, though it provides the foundation for such a system.

6. Literature Review

Research in spam detection has transitioned from simple heuristic-based methods to sophisticated machine learning and deep learning models. Naive Bayes has traditionally been a popular choice for text classification due to its simplicity and effectiveness. More recent studies have explored Support Vector Machines (SVM) and ensemble methods, which often provide

superior performance in high-dimensional text data. This project builds upon these established methodologies to create a high-performance SMS spam filter.

7. System Requirements

7.1 Hardware Requirements

- **Processor:** Intel Core i5 or higher.
- **RAM:** 8 GB (minimum).
- **Storage:** 256 GB SSD.

7.2 Software Requirements

- **Operating System:** Windows ¹⁰/₁₁ or Linux.
- **Programming Language:** Python 3.8+.
- **Key Libraries:** Pandas, NumPy, Scikit-learn, NLTK, Matplotlib, Seaborn.

8. Dataset Description

The project uses the **SMS Spam Collection** dataset, which consists of 5,169 unique SMS messages labeled as “ham” or “spam”. The dataset is imbalanced, with approximately 87% ham and 13% spam messages. This imbalance is a key consideration during model evaluation.

9. Data Cleaning and Preprocessing

Data cleaning involved removing duplicates and unnecessary columns. Text preprocessing was performed using the NLTK library:

```
# Text Preprocessing Example
def preprocess_text(text):
    text = text.lower() # Lowercasing
    text = re.sub(f"[{re.escape(string.punctuation)}]", "", text) # Punctu
    tokens = nltk.word_tokenize(text) # Tokenization
    # Stop-word removal and Stemming
    stemmer = PorterStemmer()
    tokens = [stemmer.stem(w) for w in tokens if w not in stopwords.words
    return " ".join(tokens)
```

10. Exploratory Data Analysis (EDA)

EDA revealed that spam messages tend to be longer on average than ham messages. Word clouds were generated to visualize the most frequent terms in each category, showing that spam messages often contain words like “free”, “win”, “call”, and “urgent”.

11. Feature Extraction (TF-IDF)

Term Frequency-Inverse Document Frequency (TF-IDF) was used to convert text data into numerical vectors. This technique highlights words that are unique to specific messages while downweighting common words across the dataset.

12. Machine Learning Algorithms

The project implemented and compared three models:

1. **Multinomial Naive Bayes:** Well-suited for discrete features like word counts.
2. **Logistic Regression:** A robust baseline for binary classification.
3. **Support Vector Machine (SVM):** Effective in high-dimensional spaces, particularly for text classification.

13. Model Training and Evaluation

The dataset was split into training (80%) and testing (20%) sets.

Model	Accuracy	Precision	Recall	F1-Score
Multinomial Naive Bayes	97.2%	1.00	0.78	0.88
Logistic Regression	96.1%	0.98	0.71	0.82
SVM (Linear)	98.1%	0.98	0.87	0.92

14. Results and Discussion

The **Linear SVM** model achieved the highest F1-score of **0.92**, making it the most effective model for this task. It demonstrated a strong balance between identifying spam (recall) and avoiding false alarms (precision).

15. Conclusion

The project successfully developed a high-accuracy SMS spam detection system. The results confirm that machine learning, combined with effective text preprocessing and feature extraction, can reliably filter spam messages. Future work could involve exploring deep learning models like LSTMs or Transformers for even greater accuracy.

16. References

1. NLTK Documentation: Natural Language Toolkit.
2. Scikit-learn: Text Feature Extraction.
3. Kaggle: SMS Spam Collection Dataset.